

AMARTYA KALAPAHAR

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PROFESSIONAL SUMMARY

Accomplished Software Development Engineer with a Master of Science in Computer Science from Arizona State University (2024). Proficient in software development and backend systems, with expertise in large language models, NLP, and tools like Python, PyTorch, LangChain, and OpenAI. Enhanced data processing accuracy by 25% by developing a financial analyzer using LLaMA 2, streamlining workflows and improving expense classification.

PROFESSIONAL EXPERIENCE

Self Justice

Tempe, AZ, USA

AI/ML Engineering Intern

June 2024 - Present

- Currently testing various text summarization models on my local machine using the Hugging Face library to optimize data processing techniques.
- Investigating Retrieval-Augmented Generation (RAG), LangChain, and Haystack to evaluate their potential applications in enhancing natural language processing tasks.
- Architecting scalable backend systems, streamlined data pipelines, and integrated LLMs, boosting system efficiency by 30% and cutting latency by 20%
- Working with cross-functional teams to integrate LLMs into various projects, ensuring seamless implementation and maximizing the impact of AI solutions.

Alcon

Lake Forest, CA, USA

Software Engineer Intern

May 2023 - August 2023

- Built an app using LangChain and OpenAI to interact with large text data sources, such as documentation and books, enabling efficient and accurate information retrieval and response generation.
- Crafted and deployed scalable microservices architecture, reducing system latency by 30% and improving API response times by 25%, while ensuring 99.9% uptime for high-traffic applications
- Engineered robust data pipelines to process 1M+ daily transactions, optimizing ETL workflows and reducing data processing time by 40%
- Deployed a custom AI chatbot using AWS Lambda and FastAPI, reducing hosting costs to under \$2 per 1,000 requests and increasing query processing efficiency by 40%.

Y Media Labs

Bengaluru, Karnataka, India

Software Engineer

September 2020 - June 2022

- Designed and implemented RESTful APIs for seamless integration with third-party services, increasing system interoperability and reducing development time by 15%
- Led the migration of legacy systems to cloud-based infrastructure (AWS), cutting operational costs by 20% and improving scalability for 10x user growth
- Automated CI/CD pipelines using Jenkins and Docker, reducing deployment errors by 50% and accelerating release cycles from weekly to daily
- Mentored junior developers in best practices for code reviews, testing, and debugging, improving team productivity by 20% and code quality by 30%

CVB Lab, Universitat Politècnica De Valencia

Valencia, Spain

Summer Research Intern

June 2019 - July 2019

- Programmed an AI system using Python, PyTorch, NumPy, Pandas, and ensemble learning, improving early detection rates of rare diseases by 40% and potentially saving millions in healthcare costs
- Architected and deployed optimized data storage systems utilizing PostgreSQL for transactional data and Cassandra for handling high-velocity, large-scale datasets.
- Implemented container orchestration with Kubernetes, improving deployment scalability and reducing resource costs by 25%
- Collaborated with cross-functional teams to integrate deep learning models, increasing overall model performance by 25% and securing a competitive edge in medical AI.

Pytorque Solutions Pvt. Ltd.

New Delhi, DL, India

AI Developer

December 2018 - January 2019

- Devised a text classification model using a transformer encoder combined with positional embedding, achieving a classification accuracy improvement of 15%.

- Configured an efficient training pipeline with Keras, utilizing ModelCheckpoint callbacks, which reduced training time and ensured the best model selection through rigorous validation.
- Applied advanced tokenization and embedding techniques to preprocess and prepare datasets, which improved the overall model performance by 18%, leading to more accurate classification results.
- Led the adoption of Test-Driven Development (TDD) practices, increasing test coverage by 50% and reducing post-release defects by 30%

PROJECTS

LLM-Powered Personal Finance Analyzer

December 2023 - April 2024

- Implemented a local LLaMA 2 model to securely categorize and analyze bank transactions, creating an automated financial dashboard in Python that improved expense classification accuracy by 25% and reduced manual data processing time by 30%.

AI Assistant for Economic Data Analysis

February 2024 - April 2024

- Created a custom AI assistant using OpenAI's Assistance API, integrating custom functions and a code interpreter to enhance data retrieval and analytical capabilities, and deployed it via an API endpoint, boosting user experience and functionality by 50%

KubeDataFlow

September 2023 - November 2023

- Developed a data processing pipeline using Docker, Kubernetes, Neo4j, and Kafka, boosting throughput by 40%, reducing downtime by 25%, and improving resource efficiency by 30%

Project SwiftAPI

February 2023 - March 2023

- Innovated a high-performance backend using FastAPI, crafting interactive API documentation and ensuring data integrity with Pydantic models, which significantly optimized the system's performance and developer experience.

EDUCATION

Arizona State University

August 2022 - May 2024

Master's, Computer Science

GPA: 3.77

- Relevant coursework: Fundamentals of Statistical Learning, Data Mining, Cloud Computing
- Recipient of New American Scholar Award

NMAM Institute Of Technology

August 2016 - August 2020

Bachelor's, Computer Science

GPA: 8.94

- Relevant coursework: Machine Learning, Intro to Deep Learning, Natural Language Processing

SKILLS

Programming Languages: Python, Java, SQL, C++

Frameworks and Libraries: Tensorflow, PyTorch, Keras, LangChain, OpenAI, Hugging Face, WANDB, GPT, FastAPI, Flask, Kubernetes, Kafka, Hadoop, Spark, REST APIs, Docker, LLMs, GANs, BERT, Llama, LLaMA 2, Mistral, Hugging Face Transformers, NumPy, Pandas, ChatGPT, Gemini

Tools, Databases, and OS: Jupyter Notebook, Eclipse, AWS, AWS Sagemaker, AWS EC2, AWS DynamoDB, AWS S3, ChromaDB, Neo4j, MongoDB, PostgreSQL, Unix, Linux, macOS, Visual Studio Code, Tableau, Cassandra

CERTIFICATIONS

- Udacity - Machine Learning Advanced Nanodegree
- Udacity - Machine Learning Basic Nanodegree
- Coursera - Applied Machine Learning in Python
- Coursera - Neural Networks and Deep Learning

RESEARCH PAPERS

- **Amartya Kalapahar, et al.,** "Gleason Grading of Histology Prostate Images Through Semantic Segmentation via Residual U-Net," 2020 IEEE International Conference on Image Processing (ICIP), pp. 1-5. DOI: [10.1109/ICIP40778.2020.9191250](https://doi.org/10.1109/ICIP40778.2020.9191250)